

TrES-2b

R-1514

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_140823 · created 2026-08-01T16:17:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456892.72473 +/- 0.00031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1257 +/- 0.0098
Transit depth (Rp/Rs)^2	1.58 +/- 0.25 [%]
Orbital Inclination (inc)	84.11 +/- 0.3
Airmass coefficient 1 (a1)	1.0018 +/- 0.0056
Airmass coefficient 2 (a2)	-0.001 +/- 0.01
Scatter in the residuals of the lightcurve fit is	0.56 %
Best Comparison Star	#2 - [94, 396]
Optimal Aperture	3.66
Optimal Annulus	10.05
Transit Duration (day)	0.0781 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1257 $\pm$ 0.0098 vs 0.1254 (deviation 0.03 σ)
Duration fit vs published	0.0781 d vs 0.0746 d (deviation 0.8 σ)
Mid-time O–C	0.8 min
Depth SNR	12.8
Residual scatter	0.56 %

Light curve

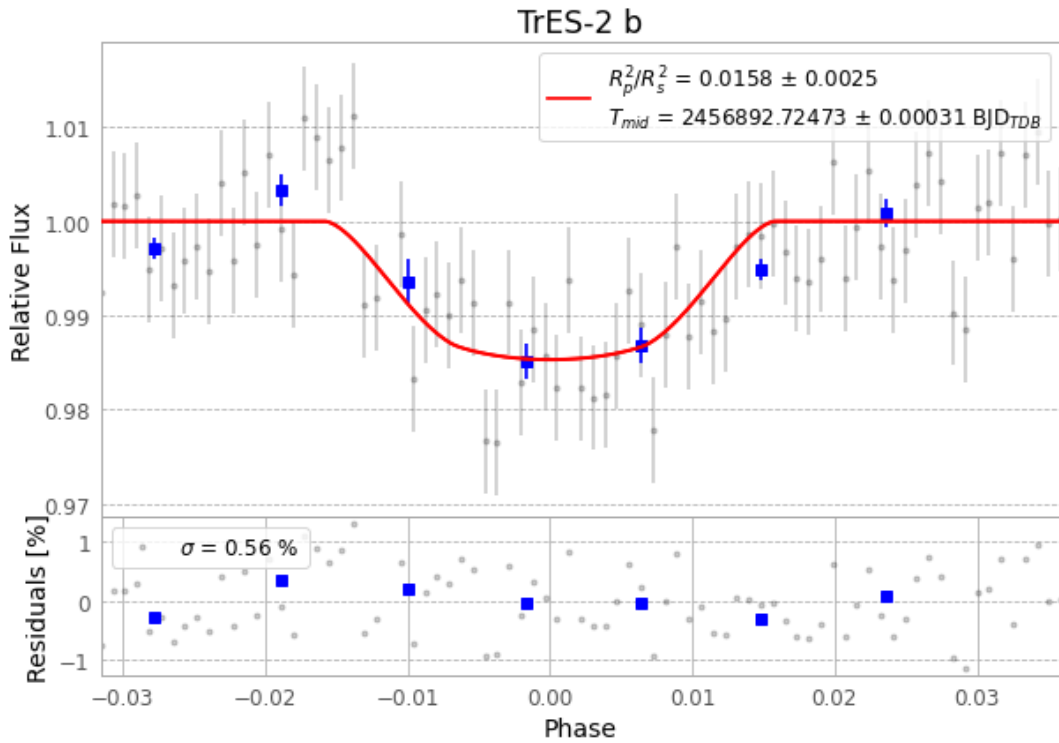


Figure 1 — FinalLightCurve_TrES-2 b_2014-08-22.png (SHA-256 d14c65bc1145ae74...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [238, 268], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ff4ca0660d70af08...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

81 FITS frames (53 MB), TRES-2140823032712.FITS → TRES-2140823072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-140823.log (350d1339d9bf...), FinalLightCurve_TrES-2 b_2014-08-22.png (d14c65bc1145...), FinalLightCurve_TrES-2 b_2014-08-22.csv (61899639fa7b...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 16:17Z.